# Ashok Singodia

**AI / ML Engineer (Internship)**

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## Summary

B.Tech Mathematics and Computing student focused on **AI/ML engineering**: **RAG pipelines**, **agentic LLM workflows**, **applied machine learning** and **computer vision**. Built three production-style systems: an **agentic hybrid graph-RAG** assistant, a **real-time fraud scoring API** and an **industrial OCR platform** at JSW Steel. Strong in **Python**, **model evaluation**, **FastAPI**, **Docker** and **CI/CD**.

## Technical Experience

**JSW Steel Ltd., Dolvi Works – IT Department Intern** 18 May 2026 – 18 Jun 2026

- **Industrial OCR System:** Built **STEELSCAN**, a full-stack **computer vision** web app that reads circular numeric coil-identification codes from steel coils, using **OpenCV** preprocessing (deskew, **circular unwarping**, contrast enhancement) with **PaddleOCR** and **EasyOCR** (Live Demo | GitHub).
- **Secure Backend:** Developed **FastAPI** REST APIs with **SQLAlchemy**, **OAuth2 and JWT** authentication, **bcrypt** hashing and **role-based access control (RBAC)**; logged each scan (timestamp, operator, confidence, source) for **traceability**; added **CSV bulk import/export** and health and metrics endpoints (request volume, success rate, latency).
- **Operator Interface:** Built a **React 18 + Vite + Tailwind** dashboard with **live camera scanning**, image upload and manual-override modes, built for plant-floor kiosks; deployed on **Vercel** and **Render**.

**YBI Foundation – AIML with Data Science Intern** 2 months, completed May 2025

- **Data Science Internship:** Completed a 2-month **AIML with Data Science** internship (specialization: **Data Science**), working on day-to-day tasks and projects using **Python**.

## Projects

**FinAdvisor-X: Agentic Hybrid Graph-RAG Financial Intelligence Platform** Live Demo | GitHub

- **Agentic Pipeline:** Built a **LangGraph** multi-agent workflow (semantic **router**, query **decomposition**, **hybrid retriever**, **math solver**, **live market data**, **Self-RAG verifier**) for grounded Q&A over **SEC 10-K filings**, a CA finance knowledge base and investment books.
- **Hybrid Retrieval:** Fused **Neo4j** vector search (**all-mpnet-base-v2**) and **Lucene** keyword search with **Reciprocal Rank Fusion (k=60)**, then reranked with a **FlashRank cross-encoder** to reduce semantic drift.
- **Hallucination Control: Self-RAG verifier** audits answers for numerical accuracy and citations and re-retrieves up to 3 times; **deterministic Python calculators** (**DCF, WACC, NPV, CAGR, SIP**, loan schedules, Indian tax rules) replace LLM arithmetic; **yfinance** adds live prices and ratios.
- **Security and Isolation: JWT** auth with **email OTP** (salted hashing, cooldown, expiry) and **bcrypt**; **per-user conversation isolation** covered by automated **pytest** tests; rate limiting and TTL caching.
- **Resilience and Deployment: Multi-provider LLM fallback** (Groq, GitHub Models, OpenRouter, Gemini); **React 19 + TypeScript** dashboard on **Vercel**, **FastAPI** backend on **Render**, **Docker**.
- **Evaluation:** Built a harness for **Precision@5**, reranker lift, router accuracy, **faithfulness** and **hallucination rate**.

**Real-Time Fraud Scoring Service (IEEE-CIS, 590K transactions)** Live Demo | GitHub

- **Modeling:** Built an end-to-end **fraud detection API** on highly imbalanced data with **LightGBM**, a strict **time-based holdout**, **isotonic probability calibration** and a **FeatureBuilder** shared by training and serving.
- **Results: 0.91 ROC-AUC**, **0.55 PR-AUC** (baseline 0.84 / 0.38), **59.7% recall at top-5% alerts**, score drift **PSI 0.0016**.
- **Cost-Based Decisions: APPROVE / MANUAL REVIEW / BLOCK** thresholds optimized on an asymmetric cost matrix ($10 review cost vs. chargeback loss); block threshold targets **90%+ precision**.
- **Explainable, Secure Serving:** Top-5 **SHAP reason codes** per prediction via **FastAPI** with **API-key auth**, **token-bucket rate limiting** and **PostgreSQL** audit logging; **Prometheus and Grafana** monitoring.
- **MLOps: Docker Compose** stack (API, PostgreSQL, Redis, Prometheus, Grafana), **GitHub Actions CI/CD**, **pytest**; deployed on **Render** with a **React + Tailwind** dashboard on **Vercel**.

## Skills

**Programming:** Python, SQL, C++, C, JavaScript, TypeScript, R, MATLAB
**ML and Deep Learning:** scikit-learn, LightGBM, XGBoost, PyTorch, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Probability Calibration, SHAP, Time-Series Forecasting, CNN, RNN, LSTM
**LLM and GenAI:** RAG, Agentic AI, LangGraph, LangChain, Hybrid Search, Reciprocal Rank Fusion, Cross-Encoder Reranking, Vector Search, Sentence-Transformers, Self-RAG, Prompt Engineering
**Computer Vision:** OpenCV, PaddleOCR, EasyOCR
**Backend and Databases:** FastAPI, REST APIs, OAuth2, JWT, RBAC, SQLAlchemy, PostgreSQL, SQLite, Redis, Neo4j (Cypher), Pandas, NumPy
**MLOps, Cloud, Frontend:** Docker, Docker Compose, GitHub Actions (CI/CD), pytest, Prometheus, Grafana, Git, Render, Vercel, React, Tailwind CSS

## Education

**Central University of Karnataka, Kalaburagi** 2023 – Expected 2027

B.Tech, Mathematics and Computing | **CGPA: 7.99** | Coursework: Machine Learning, Computational Statistics, Linear Algebra, Probability and Statistics, DBMS

## Certifications

**AI and Data Analytics Virtual Internship (Green Skills):** AICTE, Shell India Markets, Edunet Foundation, Apr–May 2025
**AI and Prompt Engineering:** VaultCodes (AICTE) | **C-DAC, Noida:** Ethical Hacking and Penetration Testing
**NPTEL:** Deep Learning; Essential Mathematics for Machine Learning; Computer Architecture